



Geospatial analysis and AHP for flood risk mapping in Quetta, Pakistan: a tool for disaster management and mitigation

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Abstract

The 2022 flood events in Quetta, Pakistan, caused severe damage to the economy, properties, and lives. Therefore, flood risk mapping to identify flood-prone areas is essential for planners and decision-makers to take critical protective measures to control the effects of flooding. This study focuses on mapping flood-prone regions in the Quetta district of Pakistan using an analytical hierarchy process (AHP) and a geographic information system (GIS). The factors influencing flood used in the present study were topographic witness index (TWI), elevation, slope, land use, land cover, precipitation, stream distance, drainage density, and soil type. Weights and ranks were allocated separately to all factors through AHP and were interpreted in a GIS environment. The produced flood hazard model of the study area depicted four zones. These zones ranged from low (19.49%), moderate (43.34%), high (28.30%), to very high (8.87%). The model was further validated through previous flood events in the study area. Around 90% of flood hazard events in the past took place mainly in the produced model's very high and high zones, which is why the current model is reliable. Finally, integrating geospatial approaches with AHP in flood hazard mapping is a quick, reliable, and affordable method that may be utilized in the area.

Keywords Flood risk mapping · Analytical hierarchy process (AHP) · RS and GIS · Balochistan · Pakistan

Introduction

Floods are among the most frequent natural disasters worldwide, resulting in significant economic, social, and human losses (Allafta and Opp 2021; Ramkar and Yadav 2021).

Over the past three decades, floods have become more severe due to human activities, which result in land use and climatic changes (Detrembleur et al. 2015; Chakraborty et al. 2022). Floods are now the most dangerous natural calamities on a global scale due to their rising frequency. However, in

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some cases, the consequences of floods are only sometimes damaging. The areas affected by a prolonged drought can benefit from a flood in terms of sustainable environmental advantages (Balica et al. 2013). According to Kim and Kim (2014), heavy rainfall is the primary cause of floods. Additionally, some human activities, including land degradation, upstream catchment deforestation, increasing housing and other property construction in flood zones (Prasad et al. 2016; Gigović et al. 2017), poor management, planning, and development of different areas, poorly managed reservoir discharges, limited capacity of drainage or stream, monsoons, and insufficient flood mitigation, have also been cited as contributing factors to floods (Danumah et al. 2016).

Natural calamities like floods have severe socioeconomic repercussions. Floods are a significant threat to the economy and livelihood of people; therefore, proper flood management is required to prevent damages caused by flood hazards. Applying specific acceptable approaches as mitigation strategies helps reduce the severity of a disaster and its aftereffects, even if it is impossible to remove the hazards connected with flood disasters (Saha and Agrawal 2020). Flood-prone area mapping is a significant technique for flood management and developing risk reduction plans, as it can produce better results. According to Cloke and Pappenberger (2009), it is almost impossible to avoid flooding altogether; however, implementing suitable methods can forecast and mitigate the effects of flood events. Identifying highly susceptible areas to flooding is the primary objective of flood-prone area mapping. To point out exposed areas before floods occur and notify locals, exploration of flood risk zones using flood risk mapping is a beneficial preventive strategy that entails researching flood-related issues (Tehrany et al. 2013). Such maps are created by considering various geo-environmental elements, such as environmental, geological, topographical, and soil properties.

Two practical tools for the management and analysis of spatial data are geographic information systems (GIS) and remote sensing (RS) (Xu et al. 2024). Since 2000, GIS-based multi-criteria analysis has not been employed for flood mapping (Kazakis et al. 2015). In recent years, numerous multi-criteria analyses have been created and used for flood hazards and risk mapping, to name some studies, frequency ratio (FR) (Samanta et al. 2018; Sahana and Patel 2019), multi-criteria decision-making (MCDM) (Ameri et al. 2018; Allafta and Opp 2021; Ramkar and Yadav 2021), analytic hierarchy process (AHP) (Kazakis et al. 2015; Gigović et al. 2017; Olii et al. 2021; Aydin and Sevgi Birincioğlu 2022), fuzzy logic (Radmehr and Araghinejad 2015; Sahana and Patel 2019). In recent years, AHP with GIS has become valuable and popular in investigating and predicting multi-parameter issues, and both are effective techniques for the spatial assessment of flood hazards (Aydin and Sevgi Birincioğlu 2022). The capacity of AHP with GIS to deal

with data scarcity is mainly responsible for the effectiveness of risk mapping (Li et al. 2024b). Topographic witness index (TWI), elevation, precipitation, stream distance, slope, soil type, drainage density, land use, and land cover were prominent influencing factors for flood-prone area mapping. Frequently, parameters are chosen through literature reviews, and their weights are allocated by professional understanding utilizing the AHP approach (Li et al. 2024a; Zhang et al. 2023).

Floods have been a very crucial and recurring natural disaster in Pakistan (Saqib et al. 2018). The Monsoon floods in 2010, 2011, and 2014 caused severe damage to basic infrastructure across the country (NDMA 2014). Similarly, the monsoon floods in 2022 left millions of Pakistanis devastated and caused huge losses to primary infrastructure and properties (Waseem and Rana 2023). Similarly, Balochistan, consisting of mountainous areas, has always been prone to flash floods (Qamer et al. 2023). Balochistan province of Pakistan received 590% additional rainfall than normal (since 1961) in August 2022 (Pakistan Meteorological Department 2022). The research studies show that both severe floods in 2010 and devastating flood in 2022 severely affected Balochistan (Cui et al. 2024).

Several research studies have been conducted on the floods, their causes, damages, and implications in Balochistan province. For instance, studies have been conducted on assessing flood damages (Hashim Nisar Hashmi 2012; Tehrany et al. 2013), crops losses, losses to infrastructure, spread and causes of flood (Cui et al. 2024), causes and implications (Nanditha et al. 2023), economic impacts of floods (Economic Survey of Pakistan 2023), assessing community preparedness and vulnerability (Jamal-Uddin and Faiz 2017; Otto et al. 2023), and assessing diseases spread during floods (Baqir et al. 2012). More precisely, like other provinces, Balochistan has also faced devastating floods, including the floods in 2022 (Zhao et al. 2024). According to the National Disaster Management Authority (NDMA) (Zhang et al. 2022), the flood of 2022 affected 32 districts out of 34, causing deaths to 336 people, injuries to 187 people, and losses of 400,000 livestock. Infrastructure damages include 214,659 houses, 58 bridges, and 2,222 km of roads (Ye et al. 2024b). Generally, heavy rainfall during the monsoon season leads to flash floods almost every year in different areas of the province (Hashim Nisar Hashmi 2012). These floods destroy people's lives, property, and livelihoods, water supply schemes, and infrastructures.

However, a recent research study has done disaster mapping of the 2022 floods in Pakistan (Wang et al. 2024). However, no study has been conducted on flood mapping in Balochistan, particularly in the Quetta district, the provincial capital of Balochistan province. Quetta district is facing several challenges, including socioeconomic and climate changes. Climate change comprises drought for long

periods, flash floods due to heavy monsoon rains, water scarcity, etc. Therefore, the current study is novel because of the following reasons. Quetta is a bowl-shaped valley surrounded by mountains (Bazai et al. 2020), which makes it highly vulnerable to flash floods. The present study area is one of the most affected districts by floods in the province. Previously, no research has been conducted on the study area to map flood risk zones through remote sensing and geographic information systems. Thus, this is the first study assessing flooding risks in Quetta, Balochistan.

Therefore, the current study attempts to map the flood risk zones in Quetta District, Balochistan, using the RS, GIS, and AHP techniques. We have utilized GIS-AHP along with eight parameters, i.e., topographic witness index (TWI), elevation, precipitation, stream distance, slope, drainage density, land use/land cover, and soil type. The present study is reliable and cost-effective, and its findings can give planners and decision-makers a more thorough examination and detailed procedures for developing pre-warning systems and emergency response actions in flooding. Further, this study also helps identify areas at greater risk due to flood hazards in the present study area.

Material and methods

Study area

Quetta is Balochistan, Pakistan's capital and most populous city. District Quetta can be found between latitudes 29°48' and 30°27' north and longitudes 66°14' and 67°18' east (Fig. 1). (Ainuddin et al. 2014). It is a remotely mountainous valley located approximately 5,500 feet above sea level with an area of 3,447 km². According to the 2017 Census, 2,269,473 people lived in the Quetta district, with a population density of 658.39 people per km² (Statistics 2017). Quetta district is very prone to earthquakes and urban floods. The Quetta district has been facing several challenges of urbanization (Mengal 2018). The unavailability of proper solid waste management measures can cause urban floods due to blockage of the sewage lines and low absorption due to the high buildup area (Zahra 2021). Urban floods are mostly flash floods that cause heavy damage due to the high slope, low absorption, and high run of mountainous topography (Luan et al. 2017). Other causes of urban floods include urbanization, unplanned development, encroachment of river channels, and improper drainage systems.

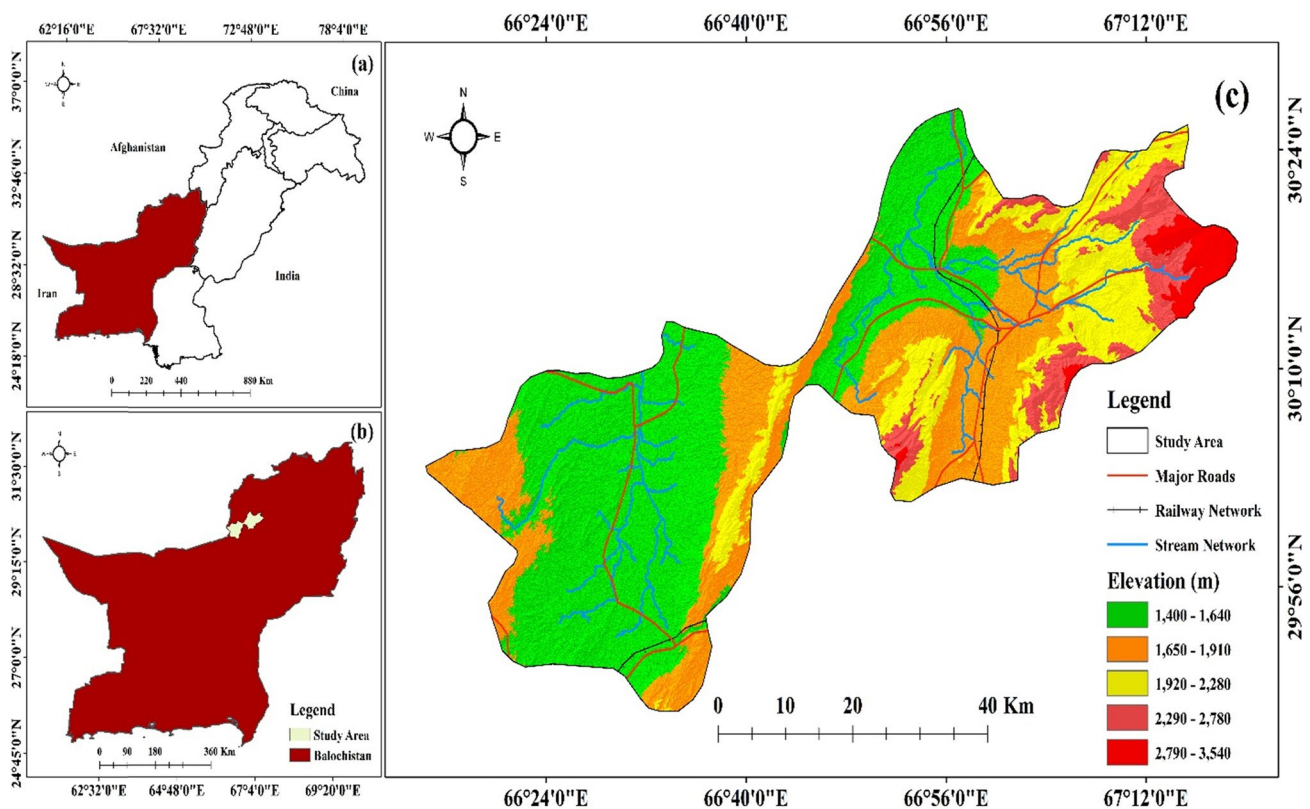


Fig. 1 Study area map, **a** Pakistan association with provincial boundaries, **b** Study area location in the province of Balochistan, **c** Study area with major roads, railway track, stream network, and elevation in meters

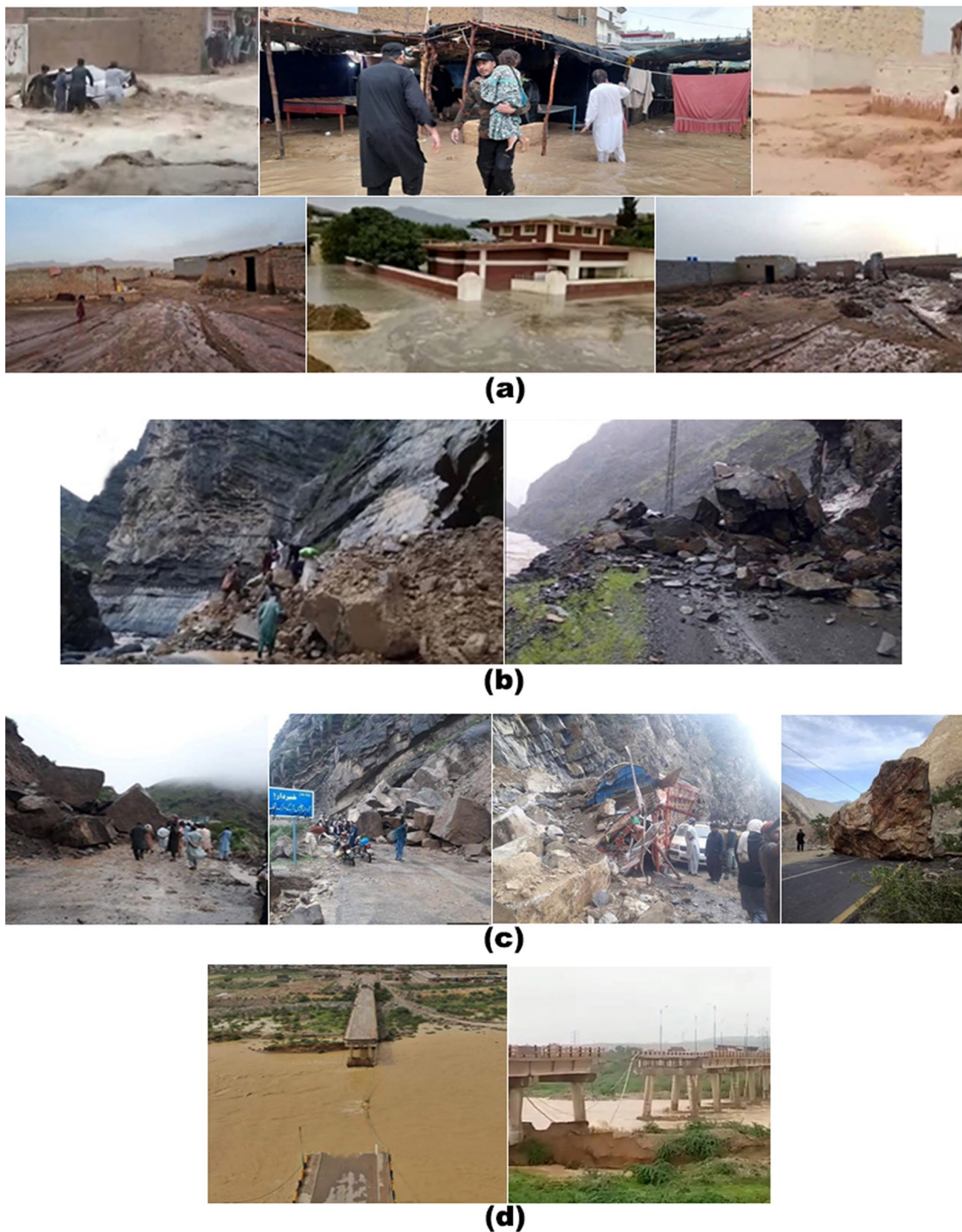


Fig. 2 Damages caused by floods in Balochistan province **a** flooded locations in the study area **b** national highway (Quetta–Peshawar) was suspended at Dhana Sar, Zhob due to land sliding **c** national highway

(Quetta–Islamabad) was suspended at Fort Monroe, Dera Ghazi Khan **d** Quetta–Karachi highway was cut off due to the breakage of the Hub Bridge

Generally, developing countries need a proper record system for flood events (Chakraborty et al. 2022). As in Pakistan, especially in Balochistan province, there is no record of hazardous events, but many urban flash flood hazard events have occurred in different years in the study area. But during 2022 in Quetta, heavy rainfall and flooding started on July 4th, 2022, affecting many areas of the city, including the Eastern and Western Bypass, Sariab, Hunna Urak, Nawa Killi, etc., causing losses to human life, property, and significant losses of animals and crops (Fig. 2a). Rainwater flooded numerous homes and made many of them unusable. Due to excessive rainfall and land sliding, Quetta's main railway line and its highways were destroyed and cut off, effectively cutting the city off from the rest of the country. The traffic on Quetta–Peshawar (N50) was suspended at Dhana Sar (Zhob) (Fig. 2b), and Quetta–Islamabad (N70) was suspended at Fort Monroe (Dera Ghazi Khan) (Fig. 2c) due to land sliding for several weeks. Along with this, the Quetta–Karachi highway was cut off due to the breakage of the Hub Bridge (Fig. 2d) by heavy floods.

Data sources and description

Different data sources were utilized to accomplish the aims and objectives of the study area. The annual rainfall data for the study area from 2010 to 2022 were collected from the irrigation department and regional meteorological center (RMC) in Quetta. SRTM digital elevation model (DEM), which has a resolution of 30 m, was obtained from the USGS website (Table 1) to calculate the study area's TWI, elevation, slope, stream network, and drainage density. The soil type toposheet with a scale of 1:2,000,000 was acquired from the Soil Survey of Pakistan to extract the study area. Additionally, for land use and land cover, Sentinel-2 images with 10 m resolution for the study area were downloaded for free from online sources (Table 1). In the (Fig. 3), we explained the procedure of flood risk assessment.

Data analysis

All factors selected for flood-prone area mapping were analyzed using the weighted overlay analysis command in a GIS environment. The analytical hierarchy process (AHP) technique has been used to choose the flood-influencing factors, assign relative scores, perform a pairwise comparison matrix, and evaluate the matrix consistency. During pairwise comparisons, the expert is asked to select a factor (e.g., TWI) for comparison with other factors one by one in identifying flood-prone areas according to Saaty's 1–9 weighing scale (Table 2). The Excel sheet automatically generates an (8 × 8) pairwise comparison matrix containing a normalized principal eigenvector for the AHP-based flood-prone area mapping.

Selection of flood-influencing parameters

To find flood-prone areas in a region, different influencing and stimulating factors and their relation to floods need to be explored (Radmehr and Araghinejad 2015; Sahana and Patel 2019). Various flood-influencing factors were utilized in earlier flood risk mapping (Kazakis et al. 2015; Saha and Agrawal 2020; Allafta and Opp 2021). Although there is no uniformity in choosing influencing parameters and their related importance, researchers choose flood-influencing parameters according to different features of a region. Based on the previous studies, we used eight flood-influencing parameters for the present study (Table 3). These parameters are topographic witness index (TWI), elevation, slope, LULC, precipitation, stream distance, drainage density, and soil type. All eight parameters were prepared in a GIS environment. Multiple institutional and remote sensing datasets were acquired for the study area's flood-prone map. All the parameters were converted into rasters and reclassified under the 'Reclassify' command in ArcGIS 10.8. under Spatial Analyst Tools.

Table 1 Detailed description of datasets, their source, and purpose

Sr. no.	Data type	Description	Spatial resolution	Temporal resolution	Source
1	(SRTM) DEM	Essential for the preparation of the TWI, elevation, slope, stream network, and drainage density	30 m	Multi-Days	USGS Earth Explorer, United States Geological Survey https://earthexplorer.usgs.gov/
2	LULC Map	Important for the analysis of land use and land cover patterns	10 m	10 days	Copernicus Data Space Ecosystem (Sentinal-2) https://dataspace.copernicus.eu/
3	Rainfall Data	High-resolution rainfall annual data for the generation of rainfall map	1 km	01 year	Regional Meteorological Center (RMC), Quetta https://rmcbalochistan.pmd.gov.pk/
4	Soil Data	Information about soil types to generate a soil type map	Scale 1:2,000,000	–	Soil Survey of Punjab https://soil.punjab.gov.pk/

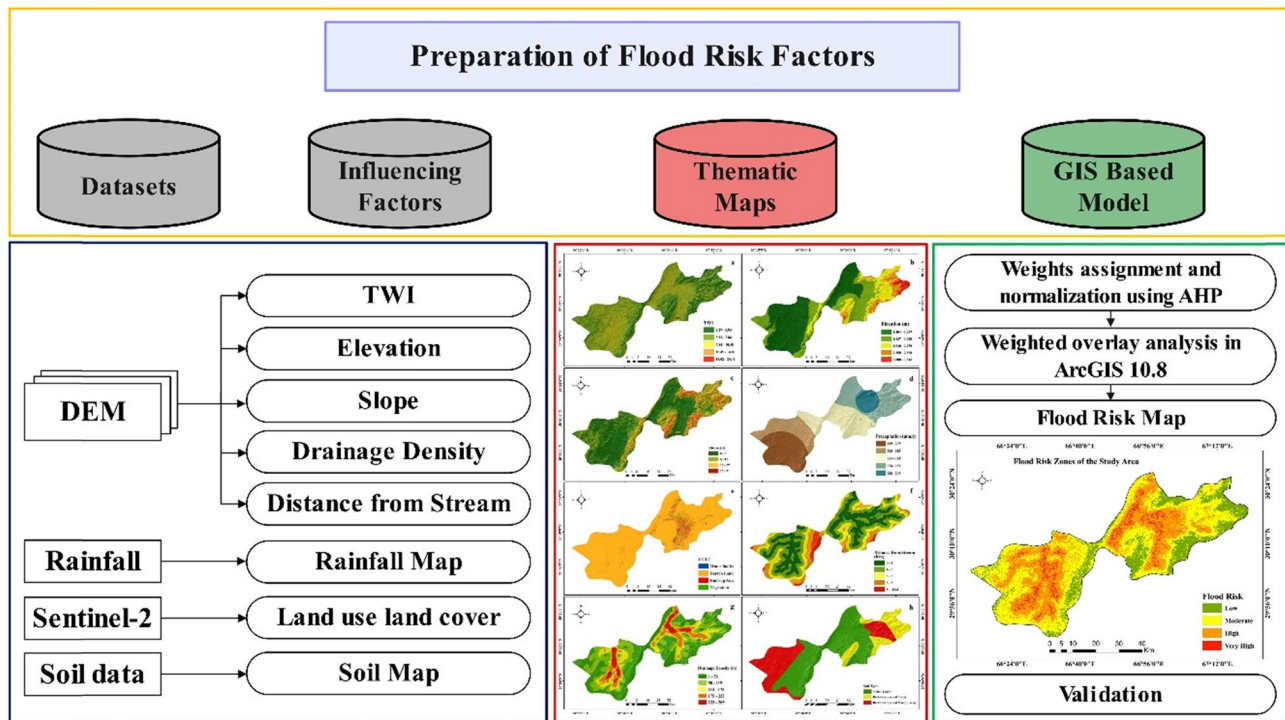


Fig. 3 Detailed flowchart shows the methodology followed for the present study

Table 2 Scales for pairwise comparison in AHP (Saaty 1990)

S. no.	Importance intensity (scores)	Definition
1	1	Equal importance
2	3	Moderate importance of one over another
3	5	The vital importance of one over another
4	7	The extreme importance of one over another
5	9	The extreme importance of one over another
6	2, 4, 6, 8	Intermediate values

Topographic wetness index (TWI)

The topographic wetness index (TWI) is a measure of surface saturation and soil moisture content (Yong et al. 2012). Flooding is stimulated by higher soil saturation and moisture content (Ho-Hagemann et al. 2015). As the saturation level increases, the groundwater table rises. Ultimately, the aeration zone becomes saturated, favoring the conditions for floods to occur. Thus, flooding is more likely to occur in areas with higher TWI.

Elevation

Elevation is measured as an important factor contributing to flood (Seejata et al. 2018). Areas with low elevation are more prone to flood than areas with high elevation (Tang et al. 2018). It also impacts the flood's runoff speed, depth, and status.

Slope

The slope is still another important factor that affects floods. Flat land or areas with low slopes are more vulnerable to flooding. A steep slope increases the runoff velocity and tends to lower the infiltration rate. So, water can swiftly go downwards to rivers as it overland flows. Thus, the angle of the slope has an impact on both the actual floods and the chances of floods.

Rainfall

The amount and season of rainfall also determine the impacts and risks of floods. Rainfall and flood risks are directly proportional to each other (Li et al. 2022). As the amount of rainfall increases, the risks of floods increase and vice versa. Furthermore, rainfall enhances the water level and strengthens the severity of surface water flooding.

Table 3 Assigned weights and rank scores for the layer/class of Quetta district

Parameter	Unit	Class	Class ratings	Class rank-ings	Parameter weight (%)
Topographic Wetness Index (TWI)	level	24.60–14.02	Very high	1	13.30
		14.02–10.4	High	2	
		10.40–7.84	Moderate	3	
		07.84–5.93	Low	4	
		05.93–< 2.27	Very low	5	
Elevation	m	1400–1637	Very high	5	12.53
		1637–1910	High	4	
		1910–2283	Moderate	3	
		2283–2782	Low	2	
		2782–3543	Very low	1	
Slop	Degree (°)	0–6	Very high	5	12.50
		6–15	Moderate	3	
		15–27	Low	2	
		27–71	Very low	1	
Rainfall	mm/y	199–205	Very low	1	14.90
		205–213	Low	2	
		213–224	Moderate	3	
		224–233	High	4	
		233–249	Very high	5	
LULC	Level	Water Bodies	Very high	5	9.92
		Vegetation	High	4	
		Built up Area	Low	2	
		Barren Area	Very low	1	
Stream distance	km	0 -1	Very high	5	13.89
		1–3	High	4	
		3–5	Moderate	3	
		5–8	Low	2	
		8–12.8	Very low	1	
Drainage density	m	0–58	Very low	1	11.76
		58–115	Low	2	
		115–175	Moderate	3	
		171–233	High	4	
		233–285	Very high	5	
Soil type	Level	Mainly Loamy	Very high	5	11.20
		Rock Outcrops and Mainly Loamy	Moderate	3	
		Rock Outcrops and Loamy	Low	2	

LULC

Land use and land cover have a significant impact on floods because they may speed up or slow down the surface flow, affecting the infiltration rate (Seejata et al. 2018). Forests and farmland have a higher infiltration rate, while built-up areas increase surface flow. Hence, built-up areas are more flood-prone than forests and bare land.

Stream distance

The distance from the stream greatly influences the distribution of floods. The distance from the stream is important in determining the flood-prone areas. Areas nearer to the main streams experience more severe floods compared to areas located relatively at a distance (Fernández and Lutz 2010).

Drainage density

Drainage density is a crucial factor causing floods in an area (Ghorbani Nejad et al. 2016). Higher drainage density is a reliable sign of a higher flow accumulation inside the basin. A more significant streamline density increases the quantity of excess runoff and, thus, increases the risk of floods.

Soil type

The soil type is one of the most crucial factors, as it greatly affects floods (Ouma and Tateishi 2014). The soil type factor made it possible to determine the infiltration rate into the soil.

Analytical hierarchy process (AHP)

GIS and AHP techniques have been popular trends in recent years. Gigović et al. (2017) all used different flood-affecting factors like topographic witness index (TWI), elevation, slope, rainfall, upper land cover class (LULC), drainage density, and more to map floods using AHP. The American theorist Thomas L. Saaty created the AHP (Saaty 1980). AHP is a well-organized and well-liked method for determining

priorities in problems involving multiple criteria for making decisions (Saaty and Vargas 1998). In the present study, the authors have utilized the AHP method by applying normalized weights to assess the flood risk map of the study area. Like (Allafta and Opp 2021), we have also used the AHP technique by choosing the flood-influencing factors, assigning relative scores, performing a pairwise comparison matrix (Table 4), and a normalized vector (Table 5), that evaluates the relative importance of eight parameters about the model's objective.

Pairwise comparison matrix

The pairwise comparison matrix table consists of eight flood risk-influencing parameters (i.e., TWI, slope, elevation, precipitation, LULC, drainage density, stream distance, and soil type). In the present study, the researcher is asked to select a factor (e.g., TWI) for comparison with other factors one by one based on the factor's necessity according to Saaty's 1–9 scale (Goepel 2013). The tables were then utilized in an MS Excel sheet developed by Goepel (Goepel 2013) (<https://bpmmsg.com/new-ahp-excel-template-with-multiple-inputs/>). The Excel sheet automatically generates an (8 × 8) pairwise

Table 4 A pairwise comparison matrix for the AHP-based flood-prone areas mapping

Matrix	TWI	EL	SL	RI	LULC	SD	DD	ST
TWI	1	1	1	1	3	1	1	1
EL	1	1	1	1	2	1	1	1
SL	1	1	1	1	3	1/2	1	1
RI	1	1	1	1	3	2	1	1
LULC	1/3	1/2	1/3	1/3	1	1/3	1	4
SD	1	1	2	1/2	3	1	1	1
DD	1	1	1	1	1	1	1	1
ST	1	1	1	1	1/4	1	1	1

TWI, topographic index; EL, elevation; SL, slope; RI, rainfall; LULC, land use/land cover; SD, stream distance; DD, drainage density; ST, soil type

Table 5 Normalized vector for flood parameters

Parameter	TWI	EL	SL	RI	LULC	SD	DD	ST	Average	Percentage
TWI	0.14	0.13	0.12	0.15	0.18	0.13	0.13	0.09	0.13	13.30
EL	0.14	0.13	0.12	0.15	0.12	0.13	0.13	0.09	0.13	12.53
SL	0.14	0.13	0.12	0.15	0.18	0.06	0.13	0.09	0.13	12.50
RI	0.14	0.13	0.12	0.15	0.18	0.26	0.13	0.09	0.15	14.90
LULC	0.05	0.07	0.04	0.05	0.06	0.04	0.13	0.36	0.10	9.92
SD	0.14	0.13	0.24	0.07	0.18	0.13	0.13	0.09	0.14	13.89
DD	0.14	0.13	0.12	0.15	0.06	0.13	0.13	0.09	0.12	11.76
ST	0.14	0.13	0.12	0.15	0.02	0.13	0.13	0.09	0.11	11.20
SUM	1	1	1	1	1	1	1	1	1	100.00

TWI, topographic index; EL, elevation; SL, slope; RI, rainfall; LULC, land use/land cover; SD, stream distance; DD, drainage density; ST, soil type

comparison matrix for the AHP-based flood-prone area mapping.

Assessing matrix consistency

According to Brunneli (2015), the highest eigenvalue ($\lambda_{\max}$) shows the matrix deviation from consistency. The pairwise matrix is reliable when the ($\lambda_{\max}$) is equivalent to or more than the number of factors examined (eight factors in the present study); otherwise, a new pairwise matrix should be created (Saaty 1980). The consistency ratio in the present study was equal to the required (10%) of the (Saaty 1980); otherwise, the assigned weights should be re-assessed to decrease inconsistency. The consistency of the whole matrix was calculated by calculating the consistency ratio (CR) as follows Eq. 1:

$$CR = \frac{CI}{RI} \quad (1)$$

where CR is the consistency ratio, CI is the consistency index, and RI is the random index. Its values are shown in Table 6 and Table 7, respectively. CI values are calculated through Eq. 2:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (2)$$

where $\lambda_{\max}$ is the maximum eigenvalue of the comparison matrix and n is the number of parameters.

Sensitivity analysis

All risk assessment models include inherent uncertainties about data and methodology (Glas et al. 2016). However, as each input data set has a different impact on the results, more precise data do not always translate into a decrease in uncertainty. Sensitivity analysis offers significant insights into the influence of the rates and weights allocated to each class and thematic layer on the final result (Fenta et al. 2015).

In the present study, we have used two methods of sensitivity, such as single-parameter sensitivity analysis (SPSA) and map removal sensitivity analysis (MRSA), to deduce the most influencing parameter in the produced flood risk map (FRM) and the influence of each class weights and rank. In the single-parameter sensitivity analysis (SPSA),

Table 7 Consistency check of proposed flood parameters

$\lambda_{\max}$	N	RI	CI	CR
8.84	8	1.41	0.11	0.08 (Yes)

$\lambda_{\max}$ the maximum eigenvalue; N, number of criteria; RI, random index; CI, consistency index; CR, consistency ratio

the effects of every thematic layer on the flood risk map are assessed. Through this method, the empirical weighting factor assigned to the thematic layers in the flood risk map is compared with the effective or real weighting factor for the same thematic layer. Flowing Eq. 3 was used to calculate the effective weighting factor for each theme layer.

$$W = \frac{PrPw}{FRM} \times 100 \quad (3)$$

where W represents the effective weight of each thematic layer, Pr is the rating and Pw is the weight of each thematic layer, and FRM is the flood risk map calculated using all the thematic layers.

The map removal sensitivity analysis (MRSA) is utilized to investigate the influence of eliminating any of the thematic layers used for the calculation of the flood risk model. Using MRS, every thematic layer is eliminated, and a new FRM is produced each time for the remaining layers atop one another. When a layer is eliminated, it is expressed in terms of the sensitivity index (SI) as given in Eq. 4:

$$SI = \frac{\left(\frac{FRM}{N}\right) \left(\frac{FRM'}{n}\right)}{FRM} \times 100 \quad (4)$$

where SI is the sensitivity index associated with elimination of a map; FRM is the flood risk map calculated through all the thematic layers; FRM' is the flood risk map by eliminating one thematic layer at a time; N and n are the number of thematic layers used to calculate FRM and FRM' respectively.

Table 6 Random index (RI) used to compute consistency

n	1	2	3	4	5	6	7	8	9	10
Random index (RI)	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

n, number of criteria; RI, random index

Results

Flood AHP hazard mapping

Floods are common natural hazards around the world, causing significant social and financial damages and life losses (Allafta and Opp 2021). Flood risk development must include flood mapping. In the present study, eight parameters—TWI, elevation, slope, LULC, precipitation, stream distance, drainage density, and soil type—were utilized to generate a flood risk map of Quetta, Pakistan. All these maps were prepared using GIS and AHP. The layers were used to establish TWI commonly used to quantify topographic control on hydrological process; elevation plays the leading role in surface runoff direction (Saha and Agrawal 2020); the slope is associated with the velocity of overland runoff and vertical percolation; more frequent and intense rainfall increases the risk of floods (Kim and Kim 2014). The LULC is related to runoff and affects the rate of infiltration in an area (Kazakis et al. 2015); distance from water bodies is closely related to flood events in the area; flooding is caused by drainage channel overflow; higher drainage causes higher surface runoff, which increases the flooding vulnerability (Subbarayan and Sivaranjani 2020); soil type is another important factor affecting surface overflow and infiltration rate in an area (Olii et al. 2021).

Generation of parameters

Topographic wetness index (TWI)

The steady-state wetness metric known as the compound topographic index (CTI), sometimes known as the topographic wetness index (TWI), is commonly used to assess topographic control in hydrological processes. There is a strong correlation between TWI, geomorphology, and the hydrographic position of a region (Saha and Agrawal 2020). This TWI evaluation can forecast saturated land surfaces that can generate overland flow. Along with this, TWI is also helpful in determining flooded areas due to overflow during heavy rainfall. The TWI layer of the study area is classified into five classes: 2.27–5.93 (22.4%), 5.93–7.84 (28.03%), 7.84–10.45 (33.0%), 10.45–14.04 (10.12%), and 14.02–24.46 (6.45%) (Fig. 4a). Topographic wetness index (TWI) is defined as Eq. 5:

$$TWI = \ln(\alpha / \tan \beta) \quad (5)$$

where α is the flow accumulation and $\tan \beta$ is a slope in radian.

Elevation

Elevation is one of the most crucial factors utilized in mapping out a region's flood risk (Rizeei et al. 2019). Runoff travels from high to low terrain, and low-elevation areas are more likely to experience more flooding (Allafta and Opp 2021). Sanyal and Lu (2006) found that low-elevated regions are more prone to floods; floods of low magnitude can occur here. Therefore, urban areas of the study area have low elevations (1400–1920 m), which are more prone to floods than the rest. The elevation of the study area was classified into five classes such as 1400–1637 (47.40%), 1637–1910 (26.03%), 1910–2283 (14.0%), 2283–2782 (7.52%), and 2782–3543 (5.05%) (Fig. 4b).

Slope

The slope is another critical parameter influencing flood risk because the surface slope is directly associated with the velocity of overland runoff and vertical percolation (Saha and Agrawal 2020). This is because areas with low slopes keep surface runoff longer than areas with high slopes. The research area's slope map is calculated in degrees and is classified into four classes such as 0°–6° (57.87%), 6°–15° (18.01%), 15°–27° (13.51%), and 27°–71° (10.61%) as shown in (Fig. 4c).

Rainfall

Rainfall is an essential factor in flooding because more frequent and intense precipitation increases the risk of floods, especially flash flooding (Kim and Kim 2014). The annual data on the rainfall for the years 2020–21 were acquired from the Regional Meteorological Center, Quetta, and the Irrigation Department, Quetta (Fig. 4d). The rainfall data of each meteorological station were interpolated using the IDW Inverse Distance Weight method in ArcGIS 10.8 to get the research area's rainfall patterns' spatial distribution. Therefore, the yearly rainfall amounts of the study area were found to range from a minimum of 199 mm/year to a maximum of 249 mm/year, and the research area was divided into five classes: < 199–205 mm/y, 205–213 mm/y, 213–224 mm/y, 224–233 mm/y, and 233–249 mm/y.

Land use land cover

Land use land cover (LULC) is also a necessary factor in flood risk mapping and is directly related to surface runoff (Gigović et al. 2017). The LULC affects the rate of infiltration and surface and groundwater interactions in an area (Kazakis et al. 2015). It has been found that whereas built-up

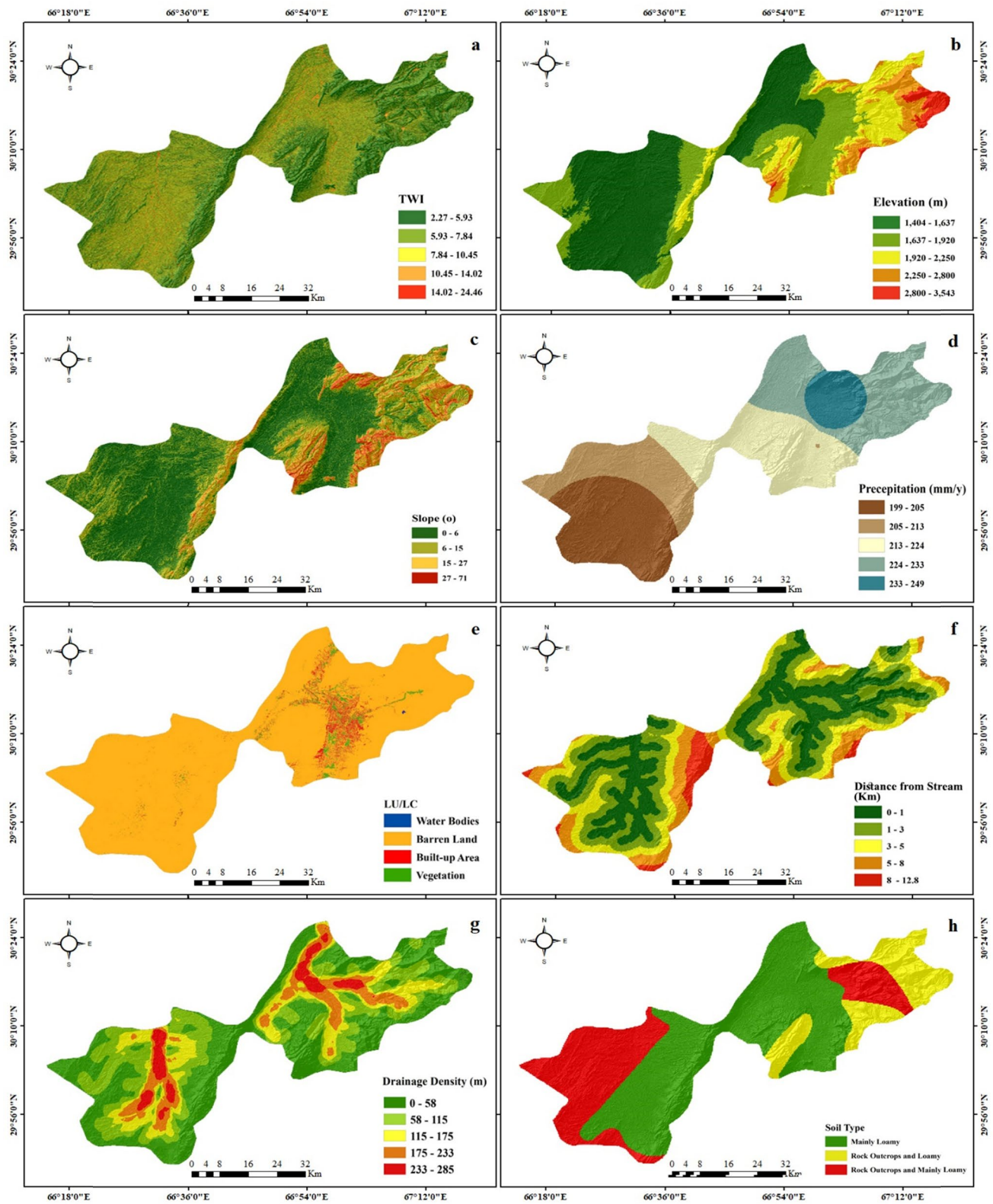


Fig. 4 Factors influencing floods: **a** TWI, **b** elevation, **c** slope, **d** rainfall, **e** LULC, **f** distance from the stream, **g** drainage density, **h** soil

areas and barren land increase surface runoff and slow infiltration rates, forests and plants can hold water for extended periods. The current research area is divided into four categories, i.e., water bodies (7.23 km²), barren land (2636.95 km²), built-up areas (541.86 km²), and vegetation (260.93 km²) (Fig. 4e).

Distance from stream

Distance from water bodies is closely related to flood events in the area; those areas that are close to water bodies, i.e., rivers, lakes, swamps, etc., are more vulnerable to flooding as compared to areas away from water bodies (Fernández and Lutz 2010). A drainage channel overflow causes flooding. A higher risk of overflow inundation exists in areas closer to drainage channels than in remote areas (Butler et al.). Therefore, the research area is classified into five categories, such as 0–1 km, 1–3 km, 3–5 km, 5–8 km, and 8–12.8 km (Fig. 4f).

Drainage density

Drainage density is another important parameter used in flood risk assessment. Flood and drainage density are

related because more drainage results in increased surface runoff, which increases the vulnerability to flooding (Subbarayan and Sivaranjani 2020). The research area's density layer is divided into five classes: 0–58 m (very low), 58–115 m (low), 115–175 m (moderate), 175–233 m (high), and 233–285 m (very high) (Fig. 4g).

Soil type

Soil type is another important factor affecting an area's surface overflow and infiltration rate (Olii et al. 2021). Surface overflow velocity typically tends to be higher, and the volume is higher in soil types containing more clay than sand. The research area's soil map, scale 1:2,000,000, was obtained from the Pakistani Soil Survey. ArcGIS 10.8 was used to digitize the map, geo-reference it, and then extract the research region using the Spatial Analyst Tools > Extract > Clip menu options. The soil map of the study area was classified based on the physical characteristics of the soil found in the research area. Three categories were created from the study area's generated soil map, i.e., mainly loamy (1758.65 km²; 51.02%), rock outcrops and loamy (1046.85 km²;

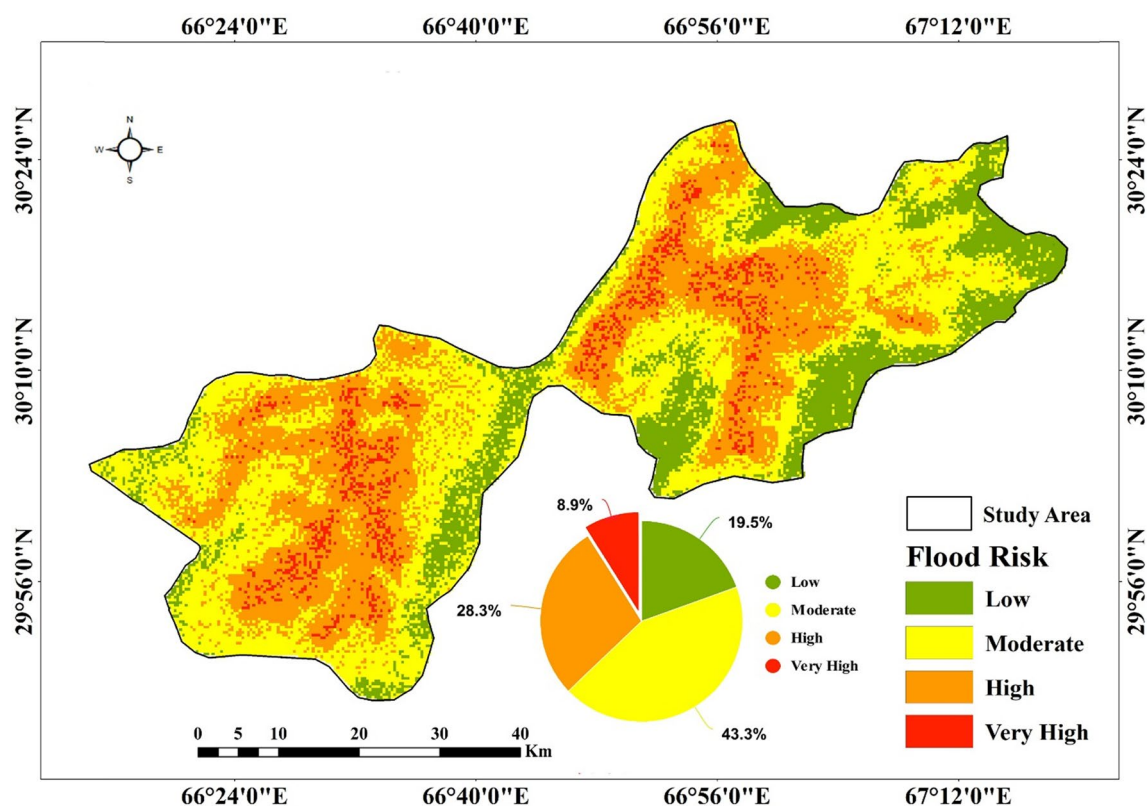


Fig. 5 Flood risk map of Quetta district, Pakistan

Table 8 Flood risk zones, area, percentage, and locations in the study area

Flood risk	Area Km ²	%	Locations in the study area
Low	671.82	19.49	Northeastern areas, Shaban Valley Areas, Chiltan Areas, Kanak, Akhtarabad, Eastern Areas
Moderate	1493.92	43.34	Panjpai, Chiltan, Kurram, Zehri Town
High	975.50	28.30	Urak, Cantonment areas, Chashma Achozai, Hazarganji, Sariyab, Muhammad Khail
Very high	305.74	8.87	Saraghurgai, Sarakhula, Pashtoon Abad, City Areas, Mariabad

30.37%), rock outcrops, and mainly loamy (641.48 km²; 18.61%) (Fig. 4h).

Flood-prone areas mapping

One of the most common natural disasters on earth's surface is flooding, which causes property loss and human life. Flood risks increase with population density, particularly in areas with high precipitation and high stream density (Aydin and Sevgi Birincioğlu 2022). For effective disaster management, flood risk assessment of an area is critical.

These days, with the support of GIS, risk analysis of natural disasters is possible, as digital backing provides efficient tools for analysis. With the assistance of the GIS environment, the current study's assessment of the study area's flood

risk was conducted using AHP. As a result, the selected eight parameters used in the study produced a flood risk map of the study area with four classes (zones): low, moderate, high, and very high (Fig. 5). About 40% of the research area is covered by zones representing places with high and very high flood risk. While elevated areas show moderate flood risk regions, high elevated areas were obtained as low flood risk regions. The study area's central portion (urban areas) is where the high (28.30%) and very high (8.87%) flood risk zones are located (Table 8). Thus, in the produced map, it has been observed that the middle parts of the study area are exposed to high flood risk. Urban areas with a high concentration of people are considered the areas at the highest risk.

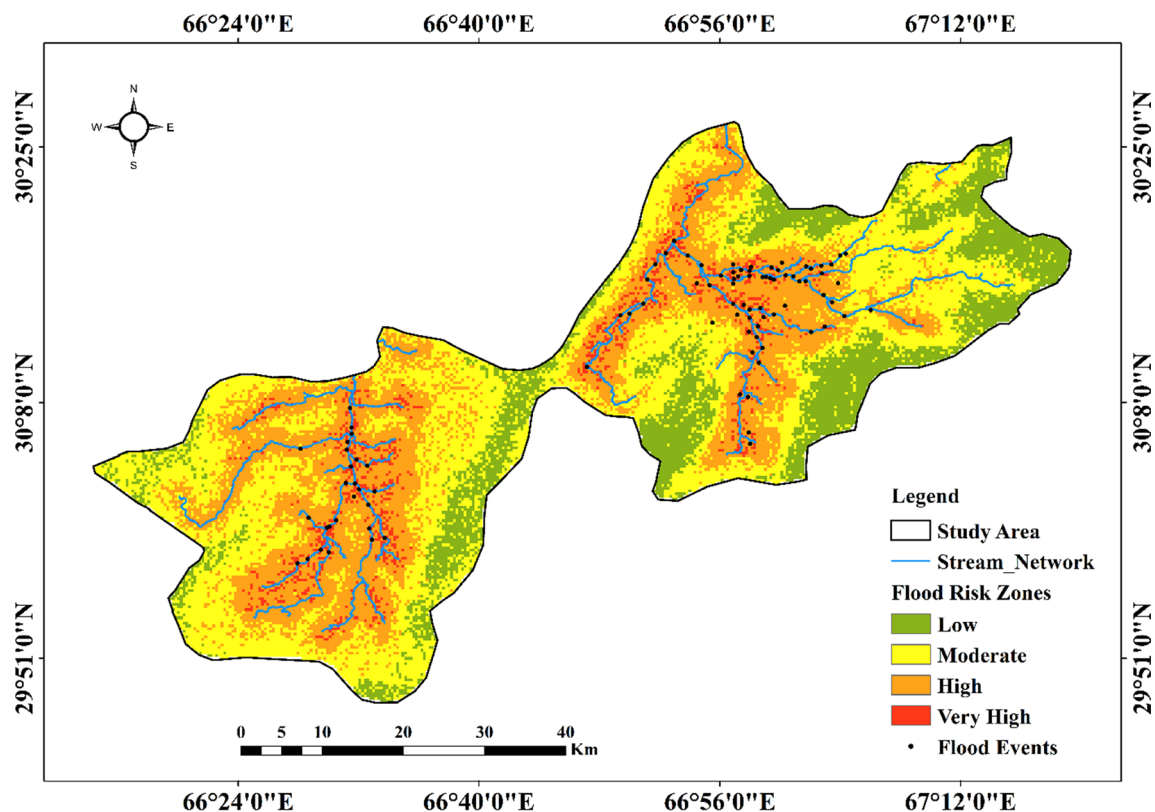
**Fig. 6** Flood risk zones and historical flood events in Quetta, Pakistan

Table 9 Statistics of single-parameter sensitivity analysis (SPSA)

Thematic layers	Empirical weight (percentage)	Effective weight (percentage)			
		Min	Max	Mean	SD
EL	34	10.01	71.25	45.23	8.12
SL	22	3.10	58.17	23.45	5.92
DD	16	2.32	47.4	12.8	5.20
SD	13	2.02	25.07	6.86	2.67
RI	6	0.7	46.0	13.7	8.2
LULC	4	0.50	10.01	2.21	1.46
TWI	3	0.44	8.18	1.30	0.88
ST	2	0.10	4.30	1.10	0.50

EL, elevation; SL, slope; DD, drainage density; SD, stream distance; RI, rainfall; LULC, land use/land cover; TWI, topographic index; ST, soil type; Min, minimum, Max, maximum, SD, standard deviation

Table 10 Statistics of map removal sensitivity analysis (MRSA)

Removed thematic layers	Variation index in percentage (%)			
	Min	Max	Mean	SD
EL	0.00	8.02	1.57	1.15
SL	0.00	5.89	1.69	0.98
DD	0.00	3.97	2.93	0.42
SD	0.00	3.74	2.22	0.79
RI	0.00	3.09	2.51	0.31
LULC	1.02	2.78	2.33	0.25
TWI	0.99	2.60	2.25	0.19
ST	0.01	2.55	1.85	0.21

EL, elevation; SL, slope; DD, drainage density; SD, stream distance; RI, rainfall; LULC, land use/land cover; TWI, topographic index; ST, soil type; Min, minimum, Max, maximum, SD, standard deviation

Flood risk model validation

Evaluating the validity and accuracy is a key task in risk management and susceptibility studies. To validate the results obtained through applying AHP, RS, and GIS, it is essential to determine the spatial relationship between the flood events that occurred in the past and the produced map of the study area. Several studies, such as [1], [20], [39], have used historical data for the validation of flood risk zone models. For the validation of the present model, we randomly selected 130 historical flood locations in the moderate-, high-, and very high-risk zones of the study area through handheld GPS (Fig. 6). From the total 130 (100%) historical flood locations, 113 (87%) of the observed historical flood events are in high and very high-risk zones of the model. Also, the produced model was compared with reports from different government organizations, i.e., National and Provincial Management Authorities (NDMA & PDMA). According to these reports, > 80% of flood events in the

past occurred mainly in high and very high-risk zones of the produced model. Therefore, the validation suggests that the produced model accurately predicted flood risk zones in the study area.

Sensitivity analysis of flood risk map (FRM)

Tables 9 and 10 comprise statistics for single-parameter sensitivity analysis (SPSA) and map removal sensitivity analysis (MRSA), respectively. When compared to the empirical weights, the single-parameter sensitivity analysis findings show variations in the effective weights of the parameters employed for flood risk mapping. The most influential parameters in flood risk mapping of the study area are elevation (EL) and slope (SL), followed by drainage density (DD). Compared to their empirical weights, stream distance (SD), rainfall (RI), land use/land (LULC), topographic index (TWI), and soil type (ST) are frequently less effective influential parameters.

According to the findings of the map removal sensitivity analysis, the flood risk map is sensitive to almost all the influencing parameters used in the present study, with maximum variation in the sensitivity index by removing drainage density (DD), and the removal of elevation (EL) causes the least variation.

Importance of using AHP-GIS

According to Paul Zwick (2007), AHP is a complete decision-making model, and its key benefit is that it can handle issues with multiple objectives and criteria by comparing each criterion's preferences in the hierarchical structure. The approaches frequently utilized in non-paired weights of GIS modeling over the past few decades have concentrated on the sequential process of calculating the weights and ranking criteria used in a model (Chang 2008). The AHP weights were computed by taking into account many pairs and computing the consistency ratio (CR) value to evaluate the weight of the logical criteria, which are occasionally disregarded in non-paired techniques due to the assumption of linearity between the criteria and their weights (Paquette and Lowry 2012).

Discussion

The risk of flooding has been extensively studied as a significant environmental hazard (Balica et al. 2012). Due to high rainfall in Asia, millions of livelihoods are severely stressed. Due to climate change, floods and other disruptions have become more common (Mukherjee et al. 2018). Additionally, due to the expected increase in air temperature and the expected extremes in rainfall during the monsoon,

the Indian subcontinent is predicted to frequently experience flooding (June–September) [54]. Floods can damage property, result in injury and death, and negatively affect a country's economy (Roy et al. 2020). Reconstruction and recovery are costly after a significant flooding disaster and might disturb a stable environment. To prevent possible floods, flood susceptibility mapping is crucial (Malik et al. 2020). Although many distinct regional geo-environmental hazard prediction methods exist, they all have the same objectives. The research community follows different techniques to create a reasonable and accurate map of vulnerability to environmental disasters. Therefore, a cost-effective and reliable model was developed in the present study to represent the distribution of flood risk zones in the Quetta District, Pakistan. An integrated AHP and GIS model was used to assess the distribution of flood hazards in the research area (Ye et al. 2024a).

In this study, a mathematical framework was built to rank decision alternatives using the GIS-based AHP approach. The AHP approach divides a complicated decision-making problem into three parts: criteria, alternatives, and a goal. Subsequently, expert opinions obtained from a consultation questionnaire are utilized to determine the relative significance of parameters through pairwise comparisons between the criteria and the alternatives of each parameter (Lyu et al. 2020). The analytical hierarchy process (AHP) has three major advantages: the hierarchical structuring to characterize the many attributes in the multi-criteria decision-making (MCDM) problems, the standardization of the relative significance, and the relatively advanced decision-making (Kai Jin and Zhang). Currently, researchers have reasonably focused on the application of AHP in flood risk assessment (Kazakis et al. 2015; Danumah et al. 2016; Subbarayan and Sivaranjani 2020; Ajay Kumar et al. 2020; Allafta and Opp 2021). These studies proved that AHP can precisely map and estimate flood risk areas. However, the results of AHP depend on the expert opinions; due to subjectivity and unpredictability, the conclusion might be intellectually constrained. For future studies, combining the AHP technique with other decision-making methodologies is recommended for more effective results.

The findings of this study have several implications for flood disaster management and mitigation in Quetta, Balochistan, and its neighboring areas. First and foremost, the present study emphasizes the need for land use planning and management to mitigate the risk of flooding. This study observed that the drainage system in Quetta is insufficient due to encroachment on drainage channels, which increases the risk of flood disaster. Therefore, investing in drainage infrastructure and its proper maintenance may reduce the effects of future floods in the study area.

In addition, the importance of the present research work rests in its practical implications. This study also helps to identify areas at greater risk due to flood hazards in the study area. The hazard models produced by the present study are valuable tools for local authorities, urban planners, and disaster management agencies, i.e., provincial and national disaster management authorities (PDMA & NDMA). Furthermore, the study findings can give decision-makers a more thorough examination and detailed procedures in different stages of the disaster management cycle by developing pre-warning systems and emergency response actions in flooding.

The findings from this study have significant implications for disaster management and mitigation in Quetta and similar regions prone to flooding. By accurately identifying flood-prone areas through integrating GIS and AHP techniques, planners and decision-makers can prioritize resource allocation and implement targeted flood mitigation strategies. The delineation of low to very high flood risk zones allows for developing tailored emergency response plans, ensuring that the most vulnerable areas receive timely assistance. Furthermore, this flood hazard model can serve as a critical tool for urban planners to enforce zoning regulations and promote resilient infrastructure development, reducing the potential for future flood damage. The methodology used in this study, which combines geospatial analysis with multi-criteria decision-making, provides a replicable and cost-effective framework that can be adapted to other regions facing similar flood risks. By incorporating real-time data and continuous monitoring, this approach can enhance flood management systems' predictive capabilities and effectiveness, ultimately safeguarding lives, property, and economic stability in flood-prone areas.

Limitations of the study

It is important to discuss a study's limitations, as they reveal possible weaknesses and restrictions in the research approach. Acknowledging a study's limitations increases the credibility of the results, helps interpret the findings more accurately, and gives directions for future research improvements.

- Precision may be compromised by input data's limited spatial resolution, e.g., SRTM 30 m, particularly for smaller-scale landscapes.
- Predictive accuracy is impacted by reliance on various input data sources, some of which may have inaccuracies or data gaps.
- During Multi-Criteria Decision Analysis (MCDA), assumptions and simplifications are made, which might lead to uncertainty, particularly when dealing with complicated interactions.

- The results of the current study could not be valid directly in other areas due to the unique geography of the Quetta Valley and the validation of the model.
- Using high-resolution input data can improve the results of the study.

Conclusion

Considering the findings of the present study in the provincial capital of Balochistan, it can be concluded that AHP and geospatial techniques provide the best and most effective results for policy-making on floods. AHP and a GIS-based study were conducted to produce a flood risk assessment map of the Quetta District, Pakistan. Eight parameters, TWI, elevation, slope, LULC, precipitation, stream distance, drainage density, and soil type, were utilized to generate a flood risk map of Quetta, Pakistan. All these maps were given weights and ranks using AHP in a GIS environment. The produced model of the study area depicted four flood risk zones: low (19.49%), moderate (43.34%), high (28.30%), and very high (8.87%). The present study's findings clearly point out that areas close to streams, with low slope angles and elevations, are at high risk of flood occurrence. The model also indicated that the urban and agricultural areas were mainly distributed in high and very high flood risk zones. Furthermore, the validation of historical flood events showed that the AHP and GIS-based models have comparatively high levels of consistency. The applied technique has several advantages, including its flexibility, cheap cost, and easy handling, that make it feasible to use, particularly in a region with limited data sources. However, using the AHP technique with other decision-making methodologies, i.e., fuzzy logic, frequency ratio, etc., is advised for more efficient results. This model is reliable for local governments and different planning, development, and flood-dealing agencies to manage and reduce flood hazard damage by devising specific flood mitigation measures in high-risk zones. The findings of this study suggest that the government needs to restore the urban drainage system, including major drains and nullahs, according to the maps of the British Era before the independence of Pakistan because only then the risk of damage from flash floods can be reduced. Secondly, clearing sludge in the sewerage system within the city can help reduce the risk of urban flooding.

Future research can incorporate real-time data and advanced machine learning techniques to improve flood risk models' accuracy and predictive capability. Exploring the socioeconomic impacts of floods and integrating these insights into risk assessments can provide a more holistic understanding of flood vulnerability. Moreover, longitudinal studies that track changes in flood patterns and the effectiveness of implemented mitigation strategies over time can

offer valuable feedback for continuous improvement in flood management practices. Collaboration with local communities to gather ground-level data and enhance community-based flood resilience measures can strengthen the overall disaster management framework.

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Author contribution Niamat Ullah, Sanaullah Panezai, Romana Ambreen, Said Qasim conceived and designed the experiments; performed the experiments; analyzed and interpreted the data; contributed reagents, materials, analysis tools, or data; wrote the paper. Aqil Tariq, Md. Galal Uddin, M. Abdullah-Al-Wadud, and Sajid Ullah contributed reagents, materials, analysis tools, or data; wrote the paper.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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References

- Ainuddin S, Kumar Routray J, Ainuddin S (2014) People's risk perception in earthquake prone Quetta city of Baluchistan. *Int J Disaster Risk Reduct* 7:165–175. <https://doi.org/10.1016/j.ijdr.2013.10.006>
- Ajay Kumar V, Mondal NC, Ahmed S (2020) Identification of ground-water potential zones using RS, GIS and AHP techniques: a case study in a part of deccan volcanic province (DVP), Maharashtra, India. *J Indian Soc Remote Sens* 48:497–511. <https://doi.org/10.1007/s12524-019-01086-3>
- Allafta H, Opp C (2021) GIS-based multi-criteria analysis for flood prone areas mapping in the trans-boundary Shatt Al-Arab basin,

- Iraq-Iran. *Geomat Nat Hazards Risk* 12:2087–2116. <https://doi.org/10.1080/19475705.2021.1955755>
- Ameri AA, Pourghasemi HR, Cerda A (2018) Erodibility prioritization of sub-watersheds using morphometric parameters analysis and its mapping: a comparison among TOPSIS, VIKOR, SAW, and CF multi-criteria decision making models. *Sci Total Environ* 613–614:1385–1400. <https://doi.org/10.1016/j.scitotenv.2017.09.210>
- Aydin MC, SevgiBirincioğlu E (2022) Flood risk analysis using GIS-based analytical hierarchy process: a case study of Bitlis Province. *Appl Water Sci* 12:122. <https://doi.org/10.1007/s13201-022-01655-x>
- Balica SF, Popescu I, Beevers L, Wright NG (2013) Parametric and physically based modelling techniques for flood risk and vulnerability assessment: a comparison. *Environ Model Softw* 41:84–92. <https://doi.org/10.1016/j.envsoft.2012.11.002>
- Balica SF, Wright NG, van der Meulen F (2012) A flood vulnerability index for coastal cities and its use in assessing climate change impacts. *Nat Hazards* 64:73–105. <https://doi.org/10.1007/s11069-012-0234-1>
- Baqir M, Sobani ZA, Bhamani A et al (2012) Infectious diseases in the aftermath of monsoon flooding in Pakistan. *Asian Pac J Trop Biomed* 2:76–79. [https://doi.org/10.1016/S2221-1691\(11\)60194-9](https://doi.org/10.1016/S2221-1691(11)60194-9)
- Bazai MH, Panzai S, Muhammad C, et al (2020) Assessment of urban sprawl and land use change dynamics through GIS and remote sensing in Quetta View project Assessment of urban sprawl and land use change dynamics through GIS and remote sensing in Quetta. *JGSS J Geogr Soc Sci* 2020:31–50
- Brunneli M (2015) Introduction to the analytic hierarchy process
- Butler D, Kokkalidou A, Makropoulos CK Supporting the siting of new urban developments for integrated urban water resource management. In: *Integrated Urban Water Resources Management*. Kluwer Academic Publishers, Dordrecht, pp 19–34
- Chakraborty R, Chandra Pal S, Rezaie F et al (2022) Flash-flood hazard susceptibility mapping in Kangsabati River Basin, India. *Geocarto Int* 37:6713–6735. <https://doi.org/10.1080/10106049.2021.1953618>
- Chang KT (2008) *Introduction to Geographic Information Systems*, vol 4. McGraw-Hill
- Cloke HL, Pappenberger F (2009) Ensemble flood forecasting: a review. *J Hydrol* 375:613–626. <https://doi.org/10.1016/j.jhydrol.2009.06.005>
- Cui L, Meng J, Li Y et al (2024) Spatiotemporal evolution characteristics of 2022 Pakistan severe flood event based on multi-source satellite gravity observations. *Remote Sens* 16:1601. <https://doi.org/10.3390/rs16091601>
- Danumah JH, Odai SN, Saley BM et al (2016) Flood risk assessment and mapping in Abidjan district using multi-criteria analysis (AHP) model and geoinformation techniques, (cote d'ivoire). *Geoenviron Dis* 3:10. <https://doi.org/10.1186/s40677-016-0044-y>
- Detrembleur S, Stilmant F, Dewals B et al (2015) Impacts of climate change on future flood damage on the river Meuse, with a distributed uncertainty analysis. *Nat Hazards* 77:1533–1549. <https://doi.org/10.1007/s11069-015-1661-6>
- Economic Survey of Pakistan (2023) *Pakistan Floods 2022 Impact Assessment* Ministry of Finance, Government of Pakistan Retrieved from
- Fenta AA, Kifle A, Gebreyohannes T, Hailu G (2015) Spatial analysis of groundwater potential using remote sensing and GIS-based multi-criteria evaluation in Raya Valley, northern Ethiopia. *Hydrogeol J* 23:195–206. <https://doi.org/10.1007/s10040-014-1198-x>
- Fernández DS, Lutz MA (2010) Urban flood hazard zoning in Tucumán Province, Argentina, using GIS and multicriteria decision analysis. *Eng Geol* 111:90–98. <https://doi.org/10.1016/j.enggeo.2009.12.006>
- Ghorbani Nejad S, Falah F, Daneshfar M, et al (2016) Delineation of groundwater potential zones using remote sensing and GIS-based data-driven models. *Geocarto Int* 1–21. <https://doi.org/10.1080/10106049.2015.1132481>
- Gigović L, Pamučar D, Bajić Z, Drobnjak S (2017) Application of GIS-interval rough AHP methodology for flood hazard mapping in urban areas. *Water* 9:360. <https://doi.org/10.3390/w9060360>
- Glas H, Deruyter G, De Maeyer P et al (2016) Analyzing the sensitivity of a flood risk assessment model towards its input data. *Nat Hazards Earth Syst Sci* 16:2529–2542. <https://doi.org/10.5194/nhess-16-2529-2016>
- Goepel KD (2013) Implementing the analytic hierarchy process as a standard method for multi-criteria decision making in corporate enterprises—a new AHP excel template with multiple inputs
- Hashim Nisar Hashmi (2012) A critical analysis of 2010 floods in Pakistan. *African J Agric Reseach* 7:1054–1067. <https://doi.org/10.5897/ajarx11.036>
- Ho-Hagemann HTM, Hagemann S, Rockel B (2015) On the role of soil moisture in the generation of heavy rainfall during the Oder flood event in July 1997. *Tellus A Dyn Meteorol Oceanogr* 67:28661. <https://doi.org/10.3402/tellusa.v67.28661>
- Jamal-Uddin SA, Faiz DGM& S (2017) Identifying factors of vulnerabilities to natural disasters: a case of malgagai refugee village Killasaifullah, Balochistan. *Bi-Annual Res J “BALOCHISTAN Rev* 37:325–336.
- Kai Jin, Zhang HC Comparison of AHP and reference point method in the environmental decision support model. In: *Conference Record 2002 IEEE International Symposium on Electronics and the Environment* (Cat. No.02CH37273). IEEE, pp 269–273
- Kazakis N, Kougias I, Patsialis T (2015) Assessment of flood hazard areas at a regional scale using an index-based approach and analytical hierarchy process: application in Rhodope-Evros region, Greece. *Sci Total Environ* 538:555–563. <https://doi.org/10.1016/j.scitotenv.2015.08.055>
- Kim B, Kim H (2014) Evaluation of flash flood severity in <sc>K</sc> area using the modified flash flood index (<sc>MFFI</sc>). *J Flood Risk Manag* 7:344–356. <https://doi.org/10.1111/jfr3.12057>
- Li Q, Song D, Yuan C, Nie W (2022) An image recognition method for the deformation area of open-pit rock slopes under variable rainfall. *Meas J Int Meas Confed* 188:110544. <https://doi.org/10.1016/j.measurement.2021.110544>
- Li F, Shan Y, Li M, et al (2024a) Insights for River Restoration: The Impacts of Vegetation Canopy Length and Canopy Discontinuity on Riverbed Evolution. *Water Resour Res* 60:e2023WR036473. <https://doi.org/10.1029/2023WR036473>
- Li J, Zhao Y, Chen D, et al (2024b) The Quantitative Role of Moisture and Vertical Motion in Shaping Summer Heavy Rainfall over North China under Two Distinct Large-Scale Weather Patterns. *J Clim* 37:2655–2672. <https://doi.org/10.1175/JCLI-D-22-0850.1>
- Luan Q, Fu X, Song C et al (2017) Runoff effect evaluation of LID through SWMM in typical mountainous, low-lying urban areas: a case study in China. *Water* 9:439. <https://doi.org/10.3390/w9060439>
- Lyu H-M, Sun W-J, Shen S-L, Zhou A-N (2020) Risk assessment using a new consulting process in fuzzy AHP. *J Constr Eng Manag*. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001757](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001757)
- Malik S, Chandra Pal S, Chowdhuri I et al (2020) Prediction of highly flood prone areas by GIS based heuristic and statistical model in a monsoon dominated region of Bengal Basin. *Remote Sens Appl Soc Environ* 19:100343. <https://doi.org/10.1016/j.rsase.2020.100343>
- Mengal S (2018) Problems of urbanization in Quetta City: an urban geography perspective. *Pakistan Geogr Rev* 73:25–34
- Mukherjee S, Aadhar S, Stone D, Mishra V (2018) Increase in extreme precipitation events under anthropogenic warming in India. *Weather Clim Extrem* 20:45–53. <https://doi.org/10.1016/j.wace.2018.03.005>

- Nanditha JS, Kushwaha AP, Singh R, et al (2023) The Pakistan Flood of August 2022: Causes and Implications. *Earth's Futur.* <https://doi.org/10.1029/2022EF003230>
- NDMA (2014) Recovery Needs Assessment and Action Framework 2014–16. Retrieved from National Disaster Management Authority. Islamabad:
- Olii MR, Olii A, Pakaya R (2021) The integrated spatial assessment of the flood hazard using AHP-GIS: the case study of gorontalo regency. *Indones J Geogr* 53. <https://doi.org/10.22146/ijg.59999>
- Otto FEL, Zachariah M, Saeed F et al (2023) Climate change increased extreme monsoon rainfall, flooding highly vulnerable communities in Pakistan. *Environ Res Clim* 2:025001. <https://doi.org/10.1088/2752-5295/acbfd5>
- Ouma YO, Tateishi R (2014) Urban flood vulnerability and risk mapping using integrated multi-parametric AHP and GIS: methodological overview and case study assessment. *Water (Switzerland)* 6:1515–1545. <https://doi.org/10.3390/w6061515>
- Pakistan Meteorological Department (2022) Pakistan's Monthly Climate Summary August, 2022 Government of Pakistan: Pakistan Meteorological Department
- Paquette J, Lowry J (2012) Flood hazard modelling and risk assessment in the Nadi River Basin, Fiji, using GIS and MCDA. *South Pacific J Nat Appl Sci* 30:33. <https://doi.org/10.1071/SP12003>
- Paul Zwick MC (2007) Smart land-use analysis: the LUCIS model. Esri incorporation
- Prasad AS, Pandey BW, Leimgruber W, Kunwar RM (2016) Mountain hazard susceptibility and livelihood security in the upper catchment area of the river Beas, Kullu Valley, Himachal Pradesh. *India Geoenviron Disasters* 3:3. <https://doi.org/10.1186/s40677-016-0037-x>
- Qamer FM, Abbas S, Ahmad B et al (2023) A framework for multi-sensor satellite data to evaluate crop production losses: the case study of 2022 Pakistan floods. *Sci Rep* 13:4240. <https://doi.org/10.1038/s41598-023-30347-y>
- Radmehr A, Araghinejad S (2015) Flood vulnerability analysis by fuzzy spatial multi criteria decision making. *Water Resour Manag* 29:4427–4445. <https://doi.org/10.1007/s11269-015-1068-x>
- Ramkar P, Yadav SM (2021) Flood risk index in data-scarce river basins using the AHP and GIS approach. *Nat Hazards* 109:1119–1140. <https://doi.org/10.1007/s11069-021-04871-x>
- Rizeei HM, Pradhan B, Saharkhiz MA (2019) Allocation of emergency response centres in response to pluvial flooding-prone demand points using integrated multiple layer perceptron and maximum coverage location problem models. *Int J Disaster Risk Reduct* 38:101205. <https://doi.org/10.1016/j.ijdrr.2019.101205>
- Roy P, Chandra Pal S, Chakraborty R et al (2020) Threats of climate and land use change on future flood susceptibility. *J Clean Prod* 272:122757. <https://doi.org/10.1016/j.jclepro.2020.122757>
- Saaty TL (1980) The analytic hierarchy process: planning, priority setting, resource allocation.
- Saaty TL (1990) How to make a decision: The analytic hierarchy process. *Eur J Oper Res* 48:9–26. [https://doi.org/10.1016/0377-2217\(90\)90057-I](https://doi.org/10.1016/0377-2217(90)90057-I)
- Saaty TL, Vargas LG (1998) Diagnosis with dependent symptoms: Bayes theorem and the analytic hierarchy process. *Oper Res* 46:491–502. <https://doi.org/10.1287/opre.46.4.491>
- Saha AK, Agrawal S (2020) Mapping and assessment of flood risk in Prayagraj district, India: a GIS and remote sensing study. *Nanotechnol Environ Eng* 5:11. <https://doi.org/10.1007/s41204-020-00073-1>
- Sahana M, Patel PP (2019) A comparison of frequency ratio and fuzzy logic models for flood susceptibility assessment of the lower Kosi River Basin in India. *Environ Earth Sci* 78:289. <https://doi.org/10.1007/s12665-019-8285-1>
- Samanta S, Pal DK, Palsamanta B (2018) Flood susceptibility analysis through remote sensing, GIS and frequency ratio model. *Appl Water Sci* 8:1–14. <https://doi.org/10.1007/s13201-018-0710-1>
- Saqib S, Kuwornu JKM, Panezia S, Ali U (2018) Factors determining subsistence farmers' access to agricultural credit in flood-prone areas of Pakistan. *Kasetsart J Soc Sci* 39:262–268. <https://doi.org/10.1016/j.kjss.2017.06.001>
- Sanyal J, Lu XX (2006) GIS-based flood hazard mapping at different administrative scales: a case study in Gangetic West Bengal, India. *Singap J Trop Geogr* 27:207–220. <https://doi.org/10.1111/j.1467-9493.2006.00254.x>
- Seejata K, Yodying A, Wongthadam T et al (2018) Assessment of flood hazard areas using Analytical Hierarchy Process over the Lower Yom Basin, Sukhothai Province. *Procedia Eng* 212:340–347. <https://doi.org/10.1016/j.proeng.2018.01.044>
- Statistics PB of (2017) 6th Population & Housing Census 2017
- Subbarayan S, Sivaranjani S (2020) Modelling of flood susceptibility based on GIS and analytical hierarchy process—a case study of Adayar River Basin. Tamilnadu, India, pp 91–110
- Tang Z, Zhang H, Yi S, Xiao Y (2018) Assessment of flood susceptible areas using spatially explicit, probabilistic multi-criteria decision analysis. *J Hydrol* 558:144–158. <https://doi.org/10.1016/j.jhydrol.2018.01.033>
- Tehrany MS, Pradhan B, Jebur MN (2013) Spatial prediction of flood susceptible areas using rule based decision tree (DT) and a novel ensemble bivariate and multivariate statistical models in GIS. *J Hydrol* 504:69–79. <https://doi.org/10.1016/j.jhydrol.2013.09.034>
- Wang J, Li K, Hao L et al (2024) Disaster mapping and assessment of Pakistan's 2022 mega-flood based on multi-source data-driven approach. *Nat Hazards* 120:3447–3466. <https://doi.org/10.1007/s11069-023-06337-8>
- Waseem HB, Rana IA (2023) Floods in Pakistan: a state-of-the-art review. *Nat Hazards Res* 3:359–373. <https://doi.org/10.1016/j.nhres.2023.06.005>
- Xu H, Zhao Y, Zhao D, et al (2024) Improvement of disastrous extreme precipitation forecasting in North China by Pangu-weather AI-driven regional WRF model. *Environ Res Lett* 19. <https://doi.org/10.1088/1748-9326/ad41f0>
- Ye B, Qiu H, Tang B et al (2024a) Creep deformation monitoring of landslides in a reservoir area. *J Hydrol* 632:130905. <https://doi.org/10.1016/j.jhydrol.2024.130905>
- Ye X, Zhu H-H, Chang F-N et al (2024b) Revisiting spatiotemporal evolution process and mechanism of a giant reservoir landslide during weather extremes. *Eng Geol* 332:107480. <https://doi.org/10.1016/j.enggeo.2024.107480>
- Yong B, Ren L-L, Hong Y et al (2012) A novel multiple flow direction algorithm for computing the topographic wetness index. *Hydrol Res* 43:135–145. <https://doi.org/10.2166/nh.2011.115>
- Zahra S (2021) Review of solid waste management of Quetta City Jan'13. 1:1–10
- Zhang K, Li Y, Yu Z et al (2022) Xin'anjiang Nested Experimental Watershed (XAJ-NEW) for Understanding Multiscale Water Cycle: Scientific Objectives and Experimental Design. *Engineering* 18:207–217. <https://doi.org/10.1016/j.eng.2021.08.026>
- Zhang J, Wang S, Huang J et al (2023) The Precipitation-Recycling Process Enhanced Extreme Precipitation in Xinjiang, China. *Geophys Res Lett* 50:e2023GL104324. <https://doi.org/10.1029/2023GL104324>
- Zhao Y, Li J, Tian Y, Li J (2024) Distinguishing Extreme Precipitation Mechanisms Associated with Atmospheric Rivers and Non-atmospheric Rivers in the Lower Yangtze River Basin. *J Clim* 37:3995–4010. <https://doi.org/10.1175/JCLI-D-23-0400.1>

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